



DC – AI – Routing – Switching – Security – Wireless – Automation

Testing BGP Roles (RFC 9234) with the JMCP Server

An HPE Juniper Networking Techpost

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HPE



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1 Testing BGP Roles (RFC 9234) with the JMCP Server

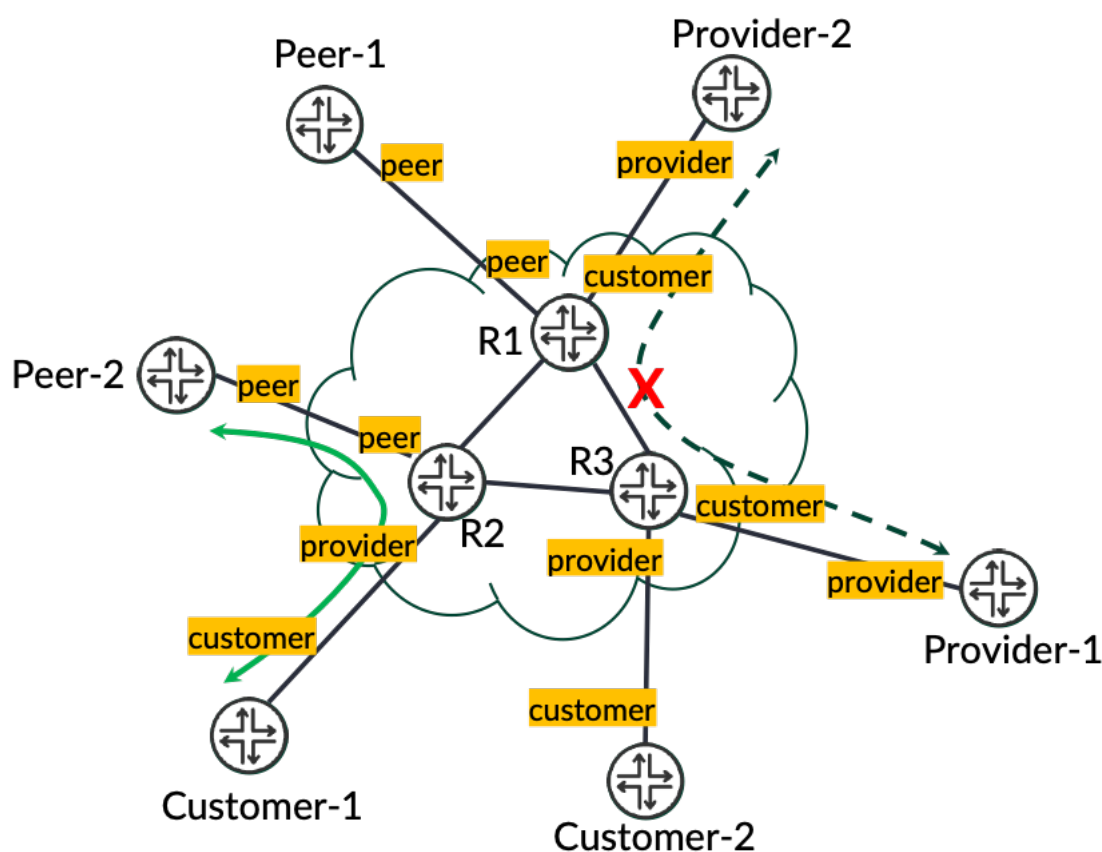
Julian Lucek - 04/22/2026

LLMs have improved significantly over the last few months with the advent of reasoning models. We are now at the point where a general-purpose LLM, in conjunction with the Junos MCP server, is capable of doing sizable chunks of work on a network triggered by a single prompt, saving the user significant amounts of time.

A valuable application of this is to help with the testing of new router features. While experimenting with new features is good fun, configuring all the pre-requisites (e.g. an IGP, the BGP sessions, AS numbers and so on) can be tedious and takes a long time, which is a barrier to getting round to testing these features for potential deployment in the production network. In this Techpost, we take the recent [BGP Roles feature \(RFC 9234\)](#) as an example, showing that the configuration of the routers (including the pre-requisites), testing, and documentation can be delegated to an LLM with a minimum of hand-holding required.

Figure 1 below illustrates how BGP Roles work. Routers R1, R2 and R3 represent an ISP that is connected to six other ASes, i.e. two customers, two peers and two upstream providers. When using BGP Roles, each side of an eBGP connection advertises in its OPEN messages its role in the context of that session. For example, router R3 advertises its role as the **customer** on its eBGP session with the upstream provider Provider-1, and Provider-1 advertises its role as the **provider**. On the other hand, router R3 advertises its role as the provider on its eBGP session with Customer-2, and Customer-2 advertises its role as the **customer**.

Another type of role is **peer**, used on sessions with lateral peers, such as the session between R2 and Peer-2. A BGP session only comes up if the roles on each side are compatible. For example, a BGP session with a role of type customer on each side would not come up.

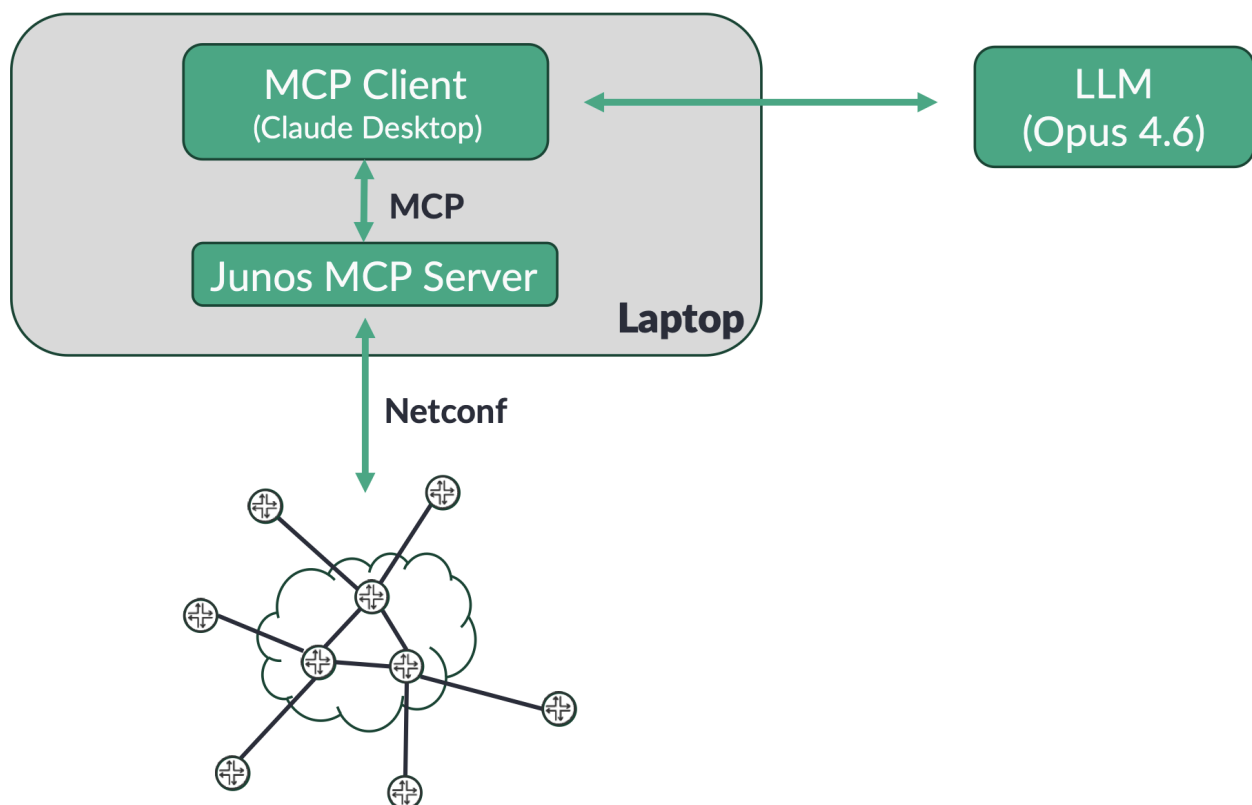


A key aspect of BGP Roles is that prefixes received from peers or providers have an Only To Customer (OTC) attribute applied. This ensures that these prefixes are only sent onwards to customers, and not to peers or providers. For example, prefixes that R3 receives from Provider-1 are propagated throughout R3's AS, but are not sent to the other provider or to the peers. They are only sent to Customer-1 and Customer-2. On the other hand, prefixes received from Customer-1 and Customer-2 do not have the OTC attribute applied and are therefore sent to all the customers, providers and peers. This removes the need for BGP communities and policies to control which eBGP neighbours receive which prefixes, making route leaks due to policy misconfiguration less likely.

In order to configure and test the network, the arrangement shown in Figure 2 was used. The MCP client was Claude Desktop with the Claude Pro licence tier. The Junos MCP server was located on the same laptop as the client. The MCP server exposes tools for the LLM to use, for example to retrieve a list of routers present in the network, retrieve router configurations, issue show commands and apply configuration to the routers.

Note that the LLM, Opus 4.6, is an off-the-shelf LLM trained by Anthropic on general internet content and other sources. As such, it has had no specialist training in networking or Junos, but presumably has been exposed to blogs and networking vendors' techpubs during the course of its general training. In addition, the LLM can trigger the retrieval of information from the internet in order to answer a user's query or perform a task if the information needed is not already within the model. In the demo shown in this article, we do not provide it any information or assistance (for example via local files, RAG, skills) apart from the queries typed into the chat window.

At the start of the demo, the network already had a basic configuration including IP and ISO addresses on the loopbacks, IP addresses on the links, SSH, login accounts and hostnames. But no routing protocol, policy or AS number configuration was present.



The first prompt entered into the chat window of the MCP Client is shown below.

- 1/ Configure ISIS between R1, R2 and R3. Point-to-point mode, Level 2 only, wide metrics only.
- 2/ Configure an AS number on each router in the network. R1, R2 and R3 are in the same AS. Each of the other routers is in its own AS. Use private AS numbers.
- 3/ Configure eBGP sessions between the AS containing R1, R2, R3 and the other ASs.
- 4/ Configure a full mesh of iBGP sessions between R1, R2 and R3.

Although I asked for quite a lot of configuration in a single prompt, the LLM successfully configured the network accordingly, as shown in the outputs below. The prompt above was a “cold start”, i.e. the LLM knew nothing about the network beforehand. After reading my prompt, the LLM started by fetching the configurations of the routers via the Junos MCP server. By looking at the subnets on each link, it worked out the topology of the network:

I'll start by examining the network topology to understand the router connections, then configure everything step by step.

Used jmcps integration >

Now I have a clear picture of the topology. Let me summarize what I've found:

Topology map (based on shared subnets):

Subnet	Interface A	Interface B	Link
17.8.0.0/24	R2 ge-0/0/1 (.1)	CUSTOMER-1 ge-0/0/1 (.2)	R2 ↔ CUSTOMER-1
17.8.1.0/24	R1 ge-0/0/1 (.1)	PROVIDER-2 ge-0/0/1 (.2)	R1 ↔ PROVIDER-2
17.8.2.0/24	R3 ge-0/0/1 (.1)	PROVIDER-1 ge-0/0/1 (.2)	R3 ↔ PROVIDER-1
17.8.3.0/24	R3 ge-0/0/2 (.1)	CUSTOMER-2 ge-0/0/1 (.2)	R3 ↔ CUSTOMER-2
17.8.4.0/24	R1 ge-0/0/2 (.1)	R2 ge-0/0/2 (.2)	R1 ↔ R2
17.8.5.0/24	R1 ge-0/0/3 (.1)	R3 ge-0/0/3 (.2)	R1 ↔ R3
17.8.6.0/24	R2 ge-0/0/3 (.1)	R3 ge-0/0/4 (.2)	R2 ↔ R3
17.8.7.0/24	R2 ge-0/0/4 (.1)	PEER-2 ge-0/0/1 (.2)	R2 ↔ PEER-2
17.8.8.0/24	R1 ge-0/0/4 (.1)	PEER-1 ge-0/0/1 (.2)	R1 ↔ PEER-1

The LLM then configured the network according to my instructions. It's interesting to note that when configuring the ISIS protocol, the LLM noticed that family iso, which is a pre-requisite for ISIS, was not configured on the links so it proactively added it. It also configured the AS numbers and the iBGP and eBGP sessions. It used loopback addresses for the iBGP sessions and link addresses for the eBGP sessions as per common practice without having to be told to do so. This effectively means that my instruction was intent-based – I

only needed to state the outcome I wanted without needing to explain the details.

Configuration Summary

1. ISIS (R1, R2, R3)

All three routers run ISIS with Level 1 disabled (Level 2 only), wide metrics only, point-to-point mode on all inter-router links, and lo0.0 as a passive interface. All 3 adjacencies (R1↔R2, R1↔R3, R2↔R3) are **Up** at Level 2. I also enabled `family iso` on the transit interfaces since it wasn't previously configured.

2. AS Number Assignments (all private)

Router	AS
R1, R2, R3	64512
CUSTOMER-1	64513
CUSTOMER-2	64514
PEER-1	64515
PEER-2	64516
PROVIDER-1	64517
PROVIDER-2	64518

3. eBGP Sessions (6 sessions, all Established)

Session	Link Subnet	Peer-AS
R1 ↔ PROVIDER-2	17.8.1.0/24	64518
R1 ↔ PEER-1	17.8.8.0/24	64515
R2 ↔ CUSTOMER-1	17.8.0.0/24	64513
R2 ↔ PEER-2	17.8.7.0/24	64516
R3 ↔ PROVIDER-1	17.8.2.0/24	64517
R3 ↔ CUSTOMER-2	17.8.3.0/24	64514

4. iBGP Full Mesh (3 sessions, all Established)

Peered using loopback addresses (reachable via ISIS): R1 (17.0.0.3) ↔ R2 (17.0.0.1), R1 (17.0.0.3) ↔ R3 (17.0.0.5), R2 (17.0.0.1) ↔ R3 (17.0.0.5).

I then typed my second prompt, to ask for some static routes to be configured and redistributed into BGP:

On each router EXCEPT R1, R2 and R3, configure a static route to discard and redistribute the routes into BGP. Check that these routes are present on each of the routers in the network, having been propagated via BGP.

The responses below show the static routes that the LLM configured on each router and a summary of the configuration steps it applied. As I had not stated what routes I wanted, the LLM proactively allocated them itself.

Configuration Summary

Static Discard Routes Configured (on 6 routers, excluding R1/R2/R3)

Router	AS	Discard Route
CUSTOMER-1	64513	192.168.13.0/24
CUSTOMER-2	64514	192.168.14.0/24
PEER-1	64515	192.168.15.0/24
PEER-2	64516	192.168.16.0/24
PROVIDER-1	64517	192.168.17.0/24
PROVIDER-2	64518	192.168.18.0/24

What was applied on each of the 6 routers:

1. **Static discard route** — `routing-options static route 192.168.x.0/24 discard`
2. **Export policy** — `policy-statement EXPORT-STATIC` matching `protocol static` → `accept`
3. **BGP export** — Applied `EXPORT-STATIC` to the eBGP group

Interestingly, the LLM noticed that it needed to add next-hop-self configuration on R1, R2 and R3, as otherwise the next-hops would have been unreachable:

Additional fix on R1, R2, R3:

The iBGP full-mesh between R1/R2/R3 lacked `next-hop-self`, causing iBGP-learned routes from eBGP peers to have unreachable next-hops (showing as "hidden/unusable"). A `next-hop-self` export policy was added to the iBGP group on all three core routers to resolve this.

Verification Results

Every router sees all **5 remote** 192.168.x.0/24 routes via BGP (each router already has its own as a local static route), with **0 hidden routes** across the entire network. The routes propagate through AS 64512 (R1/R2/R3) as expected, with AS paths like `64512 64513 I` visible on the external routers.

I was now ready to request configuration of the BGP Roles, per RFC 9234:

Configure on the BGP sessions appropriate roles, as discussed in RFC9234, Route Leak Prevention and Detection Using Roles in UPDATE and OPEN Messages.

Note that I didn't give any details about what BGP Roles are, or how to configure them, as I was hoping the LLM would work that out, which it did. It had probably been exposed to RFC 9234 during its training. On the other hand, it must have found out the corresponding Junos configuration commands via a web search, as the BGP Roles feature was released and documented after the Opus 4.6 training cut-off date. As seen in the output below, the LLM inferred from the hostnames of the routers what role to configure on each side of each eBGP session. For example, it inferred that the role of CUSTOMER-1 should be customer and therefore the role of R2 on the other end should be provider.

RFC 9234 BGP Roles — Configuration Complete

All eBGP sessions have the `BgpOTCRoleRouteLeakDetection` option enabled and roles are successfully negotiated on both sides of each session.

Negotiated Roles Per Session

Session	Local Router	Local Role	↔	Remote Router	Remote Role
R2 ↔ CUSTOMER-1	R2	provider	↔	CUSTOMER-1	customer
R3 ↔ CUSTOMER-2	R3	provider	↔	CUSTOMER-2	customer
R1 ↔ PEER-1	R1	peer	↔	PEER-1	peer
R2 ↔ PEER-2	R2	peer	↔	PEER-2	peer
R3 ↔ PROVIDER-1	R3	customer	↔	PROVIDER-1	provider
R1 ↔ PROVIDER-2	R1	customer	↔	PROVIDER-2	provider

Now that the network was fully configured, I was ready to request verification of the correct operation of BGP Roles:

Look the the RFC 9234 roles configured on the eBGP sessions on each router. Look at what prefixes are received via BGP on each of the routers (apart from from R1, R2 and R3). Provide a summary, and check if what is seen is as expected, given the role of each router on each eBGP session.

The LLM responded by providing a summary of which routers received which prefixes, followed by an analysis explaining which prefixes propagated to where and why, and showed that this was as expected from RFC 9234.

Prefixes Received via BGP on Each External Router

Router	Local Role	Received Prefixes	Count
CUSTOMER-1	customer	192.168.14/15/16/17/18.0/24	5
CUSTOMER-2	customer	192.168.13/15/16/17/18.0/24	5
PEER-1	peer	192.168.13/14.0/24	2
PEER-2	peer	192.168.13/14.0/24	2
PROVIDER-1	provider	192.168.13/14.0/24	2
PROVIDER-2	provider	192.168.13/14.0/24	2

Analysis: Is This Correct per RFC 9234?

RFC 9234 defines route propagation rules based on the Gao-Rexford model to prevent route leaks. The key principle from the **provider's (AS 64512's) perspective** is:

- **To customers:** Send all routes (customer, peer, provider-learned routes) — customers pay for full transit.
- **To peers:** Send only customer-learned routes — peers should only see your customer cone.
- **To providers:** Send only customer-learned routes — you only advertise what you're responsible for.

What AS 64512 is sending:

- **To CUSTOMER-1 and CUSTOMER-2** → 5 prefixes each (all other external prefixes) ✓
Correct. Customers receive everything — routes from other customers, peers, and providers.
- **To PEER-1 and PEER-2** → 2 prefixes each (only 192.168.13.0/24 and 192.168.14.0/24, which are the **customer** prefixes from CUSTOMER-1/AS 64513 and CUSTOMER-2/AS 64514) ✓ **Correct.** Peers only receive customer-originated routes, not routes learned from other peers or providers.
- **To PROVIDER-1 and PROVIDER-2** → 2 prefixes each (same customer prefixes: 192.168.13.0/24 and 192.168.14.0/24) ✓ **Correct.** Providers only receive customer-originated routes.

Conclusion

Everything is working as expected. The RFC 9234 role-based route leak prevention is functioning correctly. AS 64512 properly applies the valley-free routing principle: customer routes are propagated everywhere, while peer-learned and provider-learned routes are only sent to customers (downhill), never laterally to peers or uphill to providers.

Note: “recieves” was a genuine typo that I made in the chat window, I do know how to spell receives!

Now that the configuration and analysis was complete, I asked for a blog to be written about BGP Roles, based on the lab network. The prompt and the response are shown below . The blog is very good quality – the full blog is shown in Appendix A. It took the LLM about three minutes to create it, in html format.

Write a blog about how RFC 9234 works, based on the network. The audience is people who understand networking and BGP in general, but are not familiar with RFC 9234. Include a network diagram showing the role of each router and which prefix each router originates into BGP. Show which prefixes each router (apart from R1, R2 and R3) receives via BGP, and show how that is consistent with the role of each router.

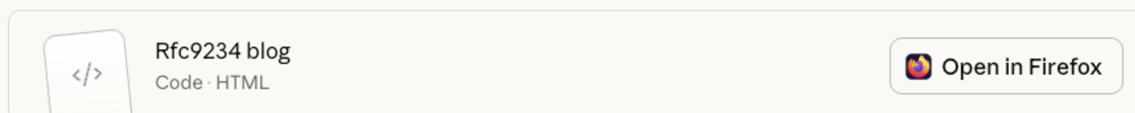
Let me read the relevant skills first.

Viewed 2 files >

Now let me create a polished, technical blog post as an HTML file with a network diagram and detailed explanation.

Created a file, ran a command, read a file >

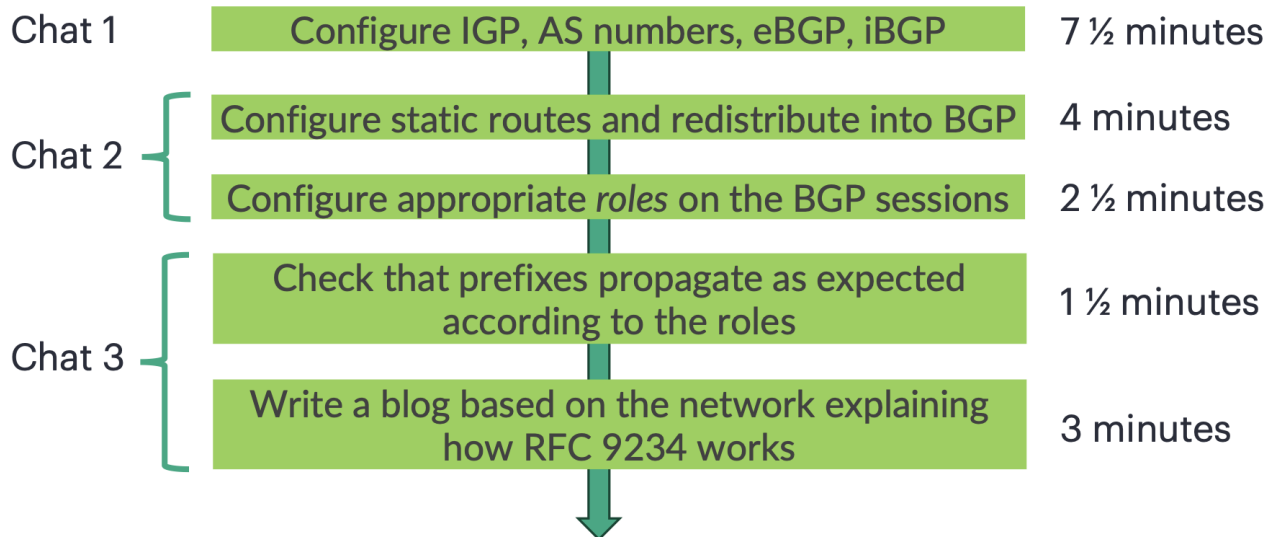
Here's the blog post. It walks through RFC 9234 using the lab network as a concrete example, covering the problem it solves, the role definitions, the valley-free propagation rules, and then shows the actual BGP table from each router as proof that the implementation matches the theory. The network diagram and colour-coded prefix tables make it easy to trace which routes flow where and why.



Some observations about the blog:

- The abstract is very good as it captures the whole essence of BGP Roles in a couple of sentences, especially where it says "...and which don't"
- The blog is well-structured, starting with a problem statement, then talking about how the RFC addresses it, then describing the lab network, then showing which prefixes propagate to where and why, followed by a discussion about the Junos configuration of the feature.
- The use of colour-coding according to the various BGP Roles is a very nice touch that makes it easy to see which prefixes propagate where.
- One flaw is that the LLM hallucinated the Junos config for BGP Roles in the blog, even though it configured this feature on the lab network correctly. This shows that LLMs are not always correct and are not always consistent.

The LLM could also be used for other types of documentation, such as internal documentation within a network operator. Figure 3 summarises the sequence of steps used in this exercise. Each green box represents one of the prompts. As shown, after the first and third prompt a new chat was started, and the old chat was discarded to avoid context window exhaustion. It might have been possible to get away with fewer but longer prompts, but I prefer to use more prompts than the bare minimum to avoid the LLM stalling half-way through a task. The duration shown for each prompt is the time from when I hit return after typing the prompt into the chat window to the time that the LLM had finished the task and reported back via the chat window.



1.1 Appendix A: blog about BGP Roles based on the lab network written by the LLM

ROUTING SECURITY

RFC 9234 in Practice: How BGP Roles Prevent Route Leaks

A walkthrough of RFC 9234's role-based route leak prevention using a real Junos lab network, showing exactly which prefixes propagate – and which don't.

The Problem: Route Leaks

If you've operated BGP at any scale, you've either experienced a route leak or read the post-mortem of one. The pattern is almost always the same: an AS receives prefixes from one provider and accidentally re-advertises them to another provider or peer, turning itself into an unintended transit network. Traffic floods through links that were never designed to carry it, latency spikes, and sometimes entire regions of the internet lose connectivity.

Historically, the only defence was careful policy configuration – prefix lists, AS-path filters, communities. But these are **unilateral**: each operator implements them independently, and there is no way for two BGP neighbours to formally agree on their commercial relationship and have the protocol itself enforce the expected route propagation rules.

This is the gap that RFC 9234 fills.

What RFC 9234 Introduces

RFC 9234 ("Route Leak Prevention and Detection Using Roles in UPDATE and OPEN Messages") adds a simple but powerful mechanism to BGP: during session establishment, both peers declare their role in the relationship. The role is carried in a new BGP OPEN message capability and is validated by both sides. If the roles are incompatible (for instance, both sides claim to be the provider), the session is rejected.

The five defined roles are:

RFC 9234 ROLE DEFINITIONS

Provider – sells transit; expects to send all routes to the customer and receive only the customer's routes in return.

Customer – buys transit; expects to receive all routes from the provider and send only its own routes upward.

Peer – settlement-free interconnection; each side sends only customer-learned routes to the other.

IS (Route Server) and RS-Client – used at IXPs; not covered in this post.

Once roles are negotiated, the router can tag outbound routes with the **Only-to-Customer (OTC)** attribute. This attribute marks a route as one that should only be propagated downhill (to customers). If a peer or provider receives a route with the OTC attribute and the route didn't come from a customer, the route is considered leaked and can be rejected.

The Lab Network

To see this in action, let's look at a lab built on Junos routers. The central AS 64512 is a transit network comprising three iBGP-meshed routers: R1, R2, and R3. Around it sit six external routers, each in its own AS, each originating a single /24 prefix into BGP.

The key detail is the RFC 9234 role configured on each eBGP session. From AS 64512's perspective, the sessions look like this:

eBGP SESSION	AS 64512 ROUTER	AS 64512'S ROLE	EXTERNAL ROUTER'S ROLE
R2 ~ CUSTOMER-1	R2	PROVIDER	CUSTOMER
R3 ~ CUSTOMER-2	R3	PROVIDER	CUSTOMER
R1 ~ PEER-1	R1	PEER	PEER
R2 ~ PEER-2	R2	PEER	PEER
R3 ~ PROVIDER-1	R3	CUSTOMER	PROVIDER
R1 ~ PROVIDER-2	R1	CUSTOMER	PROVIDER

Each external router originates a single static prefix into BGP. The prefix-to-AS mapping is straightforward: CUSTOMER-1 (AS 64513) originates 192.168.13.0/24, CUSTOMER-2 (AS 64514) originates 192.168.14.0/24, and so on through PROVIDER-2 (AS 64518) originating 192.168.18.0/24.

The Propagation Rules

RFC 9234 formalises the propagation rules that have long been considered best practice in the Geo-Berford model. From AS 64512's perspective, the rules are:

VALLEY-FREE ROUTING RULES

To customers → Send everything. Routes learned from other customers, from peers, and from providers all get advertised downhill to customers. The customer is paying for transit and should see the full routing table.

To peers → Send only customer-learned routes. A peer relationship is settlement-free, so you only share your own customer cone. You do not re-advertise routes learned from other peers or from your providers.

To providers → Send only customer-learned routes. Your provider is already giving you full transit – you should only send them routes you are authoritative for (your customers).

This is sometimes summarised as: routes only flow "downhill" from providers to customers, and "laterally" between peers – but never "uphill" from a customer or peer to a provider in a way that creates transit. The term "valley-free" comes from the fact that a valid AS path, when plotted as customer → provider relationships, should never dip down and then back up.

What the Routers Actually Receive

Now let's look at the BGP tables on each external router and see if the propagation matches the rules.

Customers: Full Table

Both CUSTOMER-1 and CUSTOMER-2 have the role `customer` on their eBGP session, meaning they expect their upstream (AS 64512) to act as a `provider` and send them everything.

ROUTER	LOCAL ROLE	PREFIXES RECEIVED	COUNT
CUSTOMER-1	CUSTOMER	192.168.14.0/24 192.168.15.0/24 192.168.16.0/24 192.168.17.0/24 192.168.13.0/24	5
CUSTOMER-2	CUSTOMER	192.168.13.0/24 192.168.15.0/24 192.168.16.0/24 192.168.14.0/24 192.168.17.0/24	5

Each customer receives **all five prefixes** originated by the other external routers – one from the other customer, two from the peers, and two from the providers. This is exactly correct: AS 64512 is acting as a transit provider and propagating its entire view of the world to its customers. / As expected

Peers: Customer Cone Only

PEER-1 and PEER-2 each have the role `peer`. They should only receive routes that AS 64512 learned from its customers.

ROUTER	LOCAL ROLE	PREFIXES RECEIVED	COUNT
PEER-1	PEER	192.168.13.0/24 192.168.14.0/24	2
PEER-2	PEER	192.168.13.0/24 192.168.14.0/24	2

Each peer receives exactly **two prefixes** – 192.168.13.0/24 from CUSTOMER-1 and 192.168.14.0/24 from CUSTOMER-2. Neither the other peer's prefix nor either provider's prefix leaks across. This is correct: in a peer relationship, only the customer cone is shared. / As expected

Providers: Customer Cone Only

PROVIDER-1 and PROVIDER-2 have the role `provider`. From AS 64512's perspective, these are upstream transit providers, and AS 64512 should only advertise its customer routes to them – never routes learned from peers or other providers.

ROUTER	LOCAL ROLE	PREFIXES RECEIVED	COUNT
PROVIDER-1	PROVIDER	192.168.13.0/24 192.168.14.0/24	2
PROVIDER-2	PROVIDER	192.168.13.0/24 192.168.14.0/24	2

Identical to the peers: only the two customer-originated prefixes are propagated upward. The peer-originated prefixes (192.168.15.0/24 and 192.168.16.0/24) and the other provider's prefix are correctly withheld. If these had leaked through, AS 64512 would be providing free transit between its providers – exactly the scenario RFC 9234 is designed to prevent. / As expected

Putting It All Together

The following table summarises the complete picture. The colour of each prefix pill indicates which type of relationship it was originally learned from – green for customer, amber for peer, purple for provider.

ROUTER	ITS ROLE	RECEIVES FROM AS 64512	VERDICT
CUSTOMER-1	CUSTOMER	14 15 16 17 13	/ All routes – correct for customer
CUSTOMER-2	CUSTOMER	13 15 16 17 14	/ All routes – correct for customer
PEER-1	PEER	13 14	/ Customer cone only – correct for peer
PEER-2	PEER	13 14	/ Customer cone only – correct for peer
PROVIDER-1	PROVIDER	13 14	/ Customer cone only – correct for provider
PROVIDER-2	PROVIDER	13 14	/ Customer cone only – correct for provider

The pattern is clear. Customer-originated routes (green) flow everywhere – they are the only routes that propagate uphill to providers and laterally to peers. Peer-originated routes (amber) and provider-originated routes (purple) only flow downhill to customers. No route leaks occur.

How Junos Implements It

In Junos, the RFC 9234 role is configured per-neighbor (or per-group) under `[protocols bgp]`. For example, on R2's session with CUSTOMER-1:

```
JUNOS CONFIGURATION EXAMPLE
The peer-as-1232 knob is set under the BGP group or neighbor. R2 configures peer-as-
role customer on its session to CUSTOMER-1, which tells Junos that R2 is the provider
and the remote AS is the customer. The role is negotiated via the OPEN message
capability, and if both sides agree, the session comes up with
bgpOTCroleRouteLeakDetection enabled – which you can see in the show bgp
neighbor output for every eBGP session in this lab.
```

Once roles are established, Junos automatically enforces the propagation rules. Routes learned from a provider or peer session are tagged with the OTC attribute, preventing them from being re-advertised to other providers or peers. No additional export policy is needed for the route leak prevention itself – it's built into the protocol.

Why This Matters

The beauty of RFC 9234 is that it shifts route leak prevention from a **configuration best-practice problem** to a **protocol-level enforcement**. Before RFC 9234, an operator could misconfigure a route-map and accidentally provide transit between two Tier-1 networks. With RFC 9234, the BGP session itself encodes the commercial relationship, and the OTC attribute provides a protocol-native mechanism for detecting and rejecting leaked routes – even if the leak happens multiple hops away.

It doesn't replace RPKI, prefix filtering, or careful BGP policy design. But it adds a powerful, low-overhead layer of defence that works within the protocol itself. If you're running Junos (or any other implementation that supports it), enabling RFC 9234 roles on your eBGP sessions is one of the highest-value, lowest-effort improvements you can make to your routing security posture.

1.2 Conclusion

In conclusion, LLMs have progressed greatly over the last few months. Previously, the type of exercise described in this article would have involved breaking down the process into smaller steps, involving a larger number of prompts and a lot more hand-holding. Here, we have seen that the LLM could handle complicated multi-step prompts, so could do large amounts of useful work correctly on its own. My prompts were effectively intent-based – I did not need to explain every step of configuration required. The LLM did all of the labwork correctly in less than sixteen minutes, which is much faster and more accurate than I would have been. It took the “initiative” to check things without being explicitly told to, for example checking that ISIS and BGP came up after it configured them. Having said that, the normal caveats when using LLMs still apply – they can make mistakes and their work needs to be checked. Nevertheless, an LLM working in conjunction with the Junos MCP Server is a very valuable tool for networking tasks.

1.3 Useful links

- <https://github.com/Juniper/junos-mcp-server>
- <https://www.juniper.net/documentation/us/en/software/junos/bgp/topics/topic-map/bgp-route-leak-prevention.html>
- <https://blog.apnic.net/2025/09/05/preventing-route-leaks-made-simple-bgp-roleplay-with-junos-rfc-9234/>
- <https://datatracker.ietf.org/doc/rfc9234/>
- <https://anthropic.com/claude-opus-4-6-system-card>, Section 1.1.1

1.4 Glossary

- BGP: Border Gateway Protocol
- IGP: Interior Gateway Protocol
- IP: Internet Protocol
- ISP: Internet Service Provider
- ISO: International Organization for Standardization
- LLM: Large Language Model
- MCP: Model Context Protocol
- NLP: Natural Language Processing
- OTC: Only To Customer
- RFC: Request for Comments
- RAG: Retrieval-Augmented Generation
- RS: Route Server
- SSH: Secure Shell



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